

Jose E. Rodríguez-Ríos

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EDUCATION

Ph.D., Software Engineering , University of Puerto Rico - Mayagüez	Aug 2026 - Present
M.S., Computer Science , University of Puerto Rico - Río Piedras GPA: 4.0/4.0	Aug 2022 - May 2025
B.S., Computer Science , University of Puerto Rico - Río Piedras Minor in Cybersecurity	Aug 2015 - Dec 2020

TECHNICAL SKILLS

Programming languages: Python, C, C++, Assembly

AI and data: PyTorch, Hugging Face, TensorFlow, LightGBM, pandas, NumPy, Jupyter, Google Colab

Vulnerability assessment: Nmap, OpenVAS, Vulners, vulscan, Nikto, Nuclei

Reverse engineering: Ghidra, BSim, GDB, strings

Forensics and analysis: Autopsy, TestDisk, Foremost, Wireshark, Scapy, steghide, binwalk, CyberChef

Security testing: Metasploit, Burp Suite, Hashcat, Hydra, Gobuster, DirBuster, John the Ripper, zip2john

Systems and DevOps: Linux, Docker, GitLab CI/CD, Git, Elasticsearch, Flask, Streamlit, SQLite, MySQL, OpenVPN

EXPERIENCE

Research Intern, Argonne National Laboratory, Lemont, IL Jun 2026 - Aug 2026

- Built DNA classifiers with **PyTorch**, **Hugging Face**, and **DNABERT** on **Perlmutter**, **Polaris**, and **Aurora**. Compared **6-mer/BPE** tokenization: **81%/80%** accuracy and **195/34 tokens** per sequence.
- Led **10 students** through a **10-12-hour** project with **Jupyter/SLURM** lessons, **CPU/GPU** experiments, and final presentations; the team fine-tuned **DNABERT** to **91.7%** evaluation accuracy on CTCF data.

Volunteer Software Developer, Abeyno Coop solar cooperative, Salinas, PR 2024 - Present

- Built **Solar Fleet Reporter**: a **1,700-line Streamlit** app with **88 automated checks** for solar data collection, net-metering calculations, and **CSV/Excel** reports and printable statements.

DevOps Intern, Lawrence Livermore National Laboratory, Livermore, CA Jun 2024 - Aug 2024

- Built a containerized **Elasticsearch** store for **Ghidra BSim** binary features so researchers can query malware-like samples at scale, with **GitLab CI/CD** for deployment.
- Deployed a **Docker**-hosted **Ollama (Llama 3)** service for the **PaperQA** PDF-reading application.

Graduate Research and Teaching Assistant, UPR Río Piedras, San Juan, PR Aug 2022 - May 2025

- Developed **NVSRCO**, an open-source vulnerability scanner using **Python**, **Nmap**, and **OpenVAS**, with **CSV** reporting, a **Dash** dashboard, and **WireGuard** links to remote networks.
- Led OS, Computer Architecture, and Cybersecurity labs with **Linux**, **Ghidra**, **Wireshark**, and **Scapy**; authored **CTF** challenges in reverse engineering, cryptography, web, and forensics for about **60** Puerto Rico high-school students.

Cybersecurity Intern, GM Sectech, San Juan, PR Jun 2022 - Aug 2022

- Built a **Python/Twilio** smishing (SMS phishing) simulator for awareness training; collaborated with penetration testers on credential-theft and domain-compromise scenarios; trained on **Metasploit**, **Hashcat**, and **Burp Suite**.

SELECTED PROJECTS

Malware Image Classification

- Built a **TensorFlow CNN** to classify **9,401** malware images across **25 families** in the **Maling** dataset.
- Reached **97.7%** validation accuracy on malware image classification.

Malicious Network Traffic Detection

- Built a **TensorFlow** pipeline for **IoT-23** logs with capture-aware splits and training-only preprocessing.
- Compared a class-weighted **dense neural network** with a **logistic-regression** baseline.

AWARDS

CAHSI Data Analytics Challenge (2026): 1st of 21 teams; 0.9428 Macro-F1; Pie Chart Pirates.

Built a **Python/LightGBM** pipeline for **36-class** network intrusion detection using **1.7M** training flows.

CAHSI Cyber Hackathon (2026): 2nd, Advanced Team Category; bacalaOS; **2,299 points**.

BSides Puerto Rico CTF (2024): 1st place of 30 teams; reverse engineering, cryptography, exploitation.

Tracer Fire Blue Team, ACSAC (2019): 2nd place team of 40 participants; SIEM/IDS, Autopsy, Ghidra.